

Probability

1. Probability measures the likelihood of an event occurring. It is defined as:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

It ranges from 0 (impossible event) to 1 (certain event).

2. Experiment: An experiment is a process or action performed to observe outcomes. It can be repeated under similar conditions.

Example: Tossing a coin, rolling a die, or drawing a card from a deck.

3. Random Experiment: A random experiment is an experiment whose outcome cannot be predicted with certainty before it is performed. It has the following characteristics:

- The set of all possible outcomes is known.
- The exact outcome of a specific trial is uncertain.

Example: Rolling a die or flipping a coin.

4. Outcome: An outcome is a single possible result of a random experiment.

Example:

- When rolling a die, an outcome could be 1, 2, 3, 4, 5, or 6.
- When flipping a coin, the outcomes are “Heads” or “Tails.”

5. A sample space S is said to be finite (infinite) sample space if the number of elements in S is finite (infinite). A sample space is called discrete if it contains only finitely or infinitely many points which can be arranged into a simple sequence $a_1, a_2, \dots$, while a sample space containing non-denumerable number of points is called a continuous sample space.

6. Trial: A trial refers to performing an experiment once.

Example: Tossing a coin once or rolling a die once.

7. Event: An event is a collection of outcomes from a random experiment. It can be:

- **Simple Event:** Consists of a single outcome. **Example:** Rolling a die and getting a 4.
- **Compound Event:** Consists of multiple outcomes. **Example:** Rolling a die and getting an even number (2, 4, 6).

8. Exhaustive Events: Exhaustive events include all possible outcomes of an experiment. The total probability of exhaustive events equals 1.

Example: In rolling a die, the exhaustive events are $\{1, 2, 3, 4, 5, 6\}$.

9. Favorable Event: A favorable event is the event that satisfies the conditions of interest.

Example: In rolling a die, the favorable event for “getting an even number” is $\{2, 4, 6\}$.

10. Mutually Exclusive Events: Two or more events are mutually exclusive if they cannot occur simultaneously.

Example:

- When tossing a coin, “Heads” and “Tails” are mutually exclusive.
- In rolling a die, the events “getting a 3” and “getting a 5” are mutually exclusive.

11. Independent Events: Two events are independent if the occurrence of one does not affect the occurrence of the other.

Example:

- Tossing two coins: The outcome of one coin does not affect the other.
- Rolling two dice: The outcome of one die does not influence the other.

Table 1: Summary of Key Terms

Term	Definition
Probability	Likelihood of an event occurring.
Experiment	A process to observe outcomes.
Random Experiment	Experiment with uncertain outcomes.
Outcome	A single possible result of an experiment.
Trial	Performing the experiment once.
Event	A set of outcomes of an experiment.
Exhaustive Events	All possible outcomes of an experiment.
Favorable Event	Outcomes that satisfy the condition of interest.
Mutually Exclusive Events	Events that cannot occur simultaneously.
Independent Events	Events where one does not influence the occurrence of the other.

Some results

- $0 \leq P(E) \leq 1$ and $P(S) = 1$.
- Probability of the impossible event is zero, i.e., $P(\emptyset) = 0$.
- Probability of the complementary event $\bar{E}$ of E is given by $P(\bar{E}) = 1 - P(A)$.
- If A and B are any two events and are not disjoint, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $P(B|A)$ represents conditional probability of occurrence of B when the event A has already happened. Then,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}.$$
- If A and B are two events, then A and B are independent if and only if $P(A \cap B) = P(A).P(B)$.

Questions

- A card is drawn from a well-shuffled pack of playing cards. What is the probability that it is either a spade or an ace?
- A box contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color. item What is the chance that a leap year selected at random will contain 53 Sundays?
- Two unbiased dice are thrown. Find the probability that: (i) both the dice shows the same number, (ii) the first die shows 6, (iii) the total of the numbers is 8, (iv) the total of the numbers is greater than 8.
- Four cards are drawn at random from a pack of 52 cards. Find the probability that (i) they are a king, a queen, a jack and an ace, (ii) two are kings and two are queens, (iii) two are black and two are red, and (iv) two are hearts and two are diamonds.
- What is the probability of getting 9 cards of the same suit in one hand at a game of bridge?
- A box contains 6 white, 4 red and 9 black balls. If 3 balls are drawn at random, find the probability that: (i) two of the balls drawn are white, (ii) one is of each color, (iii) none is red, and (iv) at least one is white.